

FHIR Integration Prospects Smartband V001

Utilizing HL7 FHIR R4 Standards for Health Data Interoperability

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What We Will Cover Today

This presentation covers the full roadmap for integrating HL7 FHIR R4 health data standards into the Smartband V001 project — from sensor hardware to hospital EHR integration.

01 What is HL7 FHIR and why it matters for wearables

05 Future sensors planned for Band01_Rev1

02 How every Smartband sensor maps to an international LOINC code

06 Five-phase implementation roadmap to full compliance

03 The four-layer FHIR system architecture

07 Data privacy, GDPR, and SMART on FHIR authentication

04 Clinical workflow — how doctors interact with FHIR data

08 Recommended next steps and procurement list

What is HL7 FHIR? — The Universal Language for Health Data

HL7 FHIR R4 (Fast Healthcare Interoperability Resources) is the international standard published by Health Level Seven International for exchanging electronic health records between any system, anywhere in the world.

The Problem FHIR Solves

Before FHIR, every hospital system, health app, and medical device used its own proprietary data format. A reading from one wearable could not be understood by a hospital's software without expensive custom integration work. FHIR replaces this chaos with a single, agreed-upon format that every modern health system speaks natively.

How it Works

FHIR breaks health data into modular objects called Resources. Each resource represents one clinical concept — a patient, a measurement, a diagnosis. Resources are transmitted as JSON over standard HTTP (REST API), making them readable by any programming language or platform.

Key FHIR Resources Used in Smartband V001

Patient

Stores the wearer's demographics, age, weight, height, and device assignment

Observation

Every single sensor reading — HR, temperature, SpO2, acceleration — as a structured data object

Bundle

A container that groups multiple Observations together into one API response

Condition

Records a detected health condition or alert — e.g. elevated heart rate episode

CarePlan

The doctor's monitoring plan — thresholds, instructions, follow-up schedule

Why FHIR for Smartband V001? — Six Key Reasons

Without FHIR, the Smartband's data is a proprietary island. With FHIR, it becomes part of the global health data ecosystem — readable by any hospital, app, or research platform in the world.



Interoperability

Works with any EHR or health app worldwide

FHIR-compliant data can be consumed by Epic, Cerner, OpenMRS, and thousands of other systems without writing a single line of custom integration code. One standard, infinite connections.



Clinical Validity

LOINC + UCUM ensure correct interpretation

Every sensor reading carries an internationally agreed code (LOINC) and unit (UCUM). A heart rate reading from Smartband V001 means exactly the same thing to a doctor in Germany as it does in Japan or the USA.



Research Ready

Direct import into research platforms

FHIR-formatted data can be loaded directly into clinical research databases like the N3C COVID Cohort or hospital analytics platforms, enabling Smartband data to contribute to real medical studies.



Regulatory Alignment

EU MDR and FDA increasingly require FHIR

The EU Medical Device Regulation (MDR 2017/745) and the US FDA both increasingly require or strongly recommend FHIR compliance for connected medical devices. Building it now prevents costly retrofits later.



EHR Integration

Connects to Epic, Cerner, OpenMRS directly

Because FHIR is the API standard for modern EHRs, Smartband V001 data can flow directly into a patient's hospital record — giving clinicians a complete longitudinal health picture that includes wearable data.



Clinical Workflow

Enables structured doctor-patient workflows

FHIR enables doctor dashboards, patient portals, structured care plans, and subscription-based real-time alerts — turning raw sensor data into a clinical-grade remote monitoring system.

Sensor → FHIR Mapping — LOINC Codes & UCUM Units

LOINC (Logical Observation Identifiers Names and Codes) is a universal database of codes for medical measurements. UCUM standardizes the units. Together they ensure that a heart rate of 75 bpm from Smartband V001 is understood identically by every hospital system globally.

What is a LOINC Code?

A LOINC code is a unique number assigned to a specific clinical measurement. For example, 8867-4 always means Heart Rate, measured in beats per minute, anywhere in the world. It is maintained by the Regenstrief Institute and used in over 168 countries.

What is UCUM?

UCUM (Unified Code for Units of Measure) standardizes how units are written. It ensures '/min' always means beats per minute, 'Cel' always means degrees Celsius, and '[g]' always means G-force — removing any ambiguity when data crosses system boundaries.

Example FHIR Observation for Heart Rate:

```
{ "resourceType": "Observation",
  "code": { "coding": [{ "system": "http://loinc.org",
    "code": "8867-4", "display": "Heart rate" }] },
  "valueQuantity": { "value": 75, "unit": "/min" } }
```

Measurement	Sensor	LOINC	Unit	Status
Heart Rate	MAX30102	8867-4	/min	On PCB ✓
SpO2	MAX30102	2708-6	%	On PCB ✓
Body Temperature	MLX90632	8310-5	Cel	On PCB ✓
Acceleration	MPU-6050	80493-0	[g]	On PCB ✓
Step Count	MPU-6050	55423-8	steps	Calculated
Calories	Calculated	41981-2	kcal	Calculated
Stress Level	GSR/EDA	74767-6	score	Future HW
Blood Pressure	PPG/Cuff	55284-4	mm[Hg]	Future HW
ECG	AD8232	11524-6	mV	Future HW

Four-Layer FHIR Architecture — From Sensor to Clinician

The Smartband V001 FHIR architecture has four distinct layers. The critical transformation happens at Layer 3 — this is where raw sensor numbers become internationally standardized FHIR resources that any hospital system can read directly.



FHIR REST API — What the Gateway Exposes

The Raspberry Pi 5 gateway acts as a FHIR server. Any device or software that speaks HTTP can call these endpoints and receive properly structured FHIR R4 resources in return — no special driver or SDK required.

How the API Works

A client sends an HTTP GET request to an endpoint. The gateway reads the latest sensor data from the CSV log, wraps it in a FHIR resource with LOINC codes and UCUM units, and returns it as JSON. The client can be a phone app, a hospital EHR, or a research script — they all get the same standardized response.

GET

/api/fhir/patient

All registered patients — demographics, device, weight, height

GET

/api/fhir/observation

Complete snapshot — all current vital observations grouped together

GET

/api/fhir/observation/hr

Heart Rate (LOINC 8867-4) — beats per minute

GET

/api/fhir/observation/temp

Body Temperature (LOINC 8310-5) — degrees Celsius

GET

/api/fhir/observation/spo2

Blood Oxygen Saturation (LOINC 2708-6) — percentage

GET

/api/fhir/observation/steps

Step Count (LOINC 55423-8) — derived from MPU-6050 motion data

GET

/api/fhir/condition

Active health alerts — e.g. tachycardia episode, elevated temperature

GET

/api/fhir/careplan

Doctor's monitoring plan — thresholds, frequency, instructions

Clinical Workflow — How Doctors Interact with Smartband FHIR Data

FHIR transforms the Smartband from a simple data collector into a clinical tool. The workflow below shows how a doctor can monitor a patient remotely, receive alerts, and document clinical decisions — all through the FHIR standard.

1

Patient wears Smartband V001

The device continuously collects HR, SpO2, body temperature, and acceleration every 2 seconds. The patient goes about their daily life wearing the band on the wrist.



2

Bluetooth transmission to gateway

The CC2640R2F firmware transmits readings via BLE GATT profile to the Raspberry Pi 5 gateway. The Pmod BT2 module on the Pi receives the data via UART at 115200 baud.



3

FHIR structuring at the gateway

The gateway maps raw sensor values to FHIR R4 Observation resources — adding LOINC codes, UCUM units, ISO 8601 timestamps, and a reference to the Patient resource.

4

Doctor accesses live FHIR data

The clinician opens the web app or their hospital EHR system. Both query the FHIR REST API and display live vitals, trend graphs, and the full alert history for each patient.



5

Threshold crossed — alert generated

When HR exceeds 100 BPM, temperature exceeds 38.5°C, or acceleration exceeds 2.0G, a FHIR Condition resource is created and a FHIR Subscription pushes an alert to the doctor's device in real time.



6

Doctor updates patient status

The doctor taps the patient card, changes status to 'Under Observation' or 'Critical', and types a clinical note. The note is stored as a FHIR Condition and visible immediately on all connected systems.

Current PCB Sensors & Planned Future Hardware

Band01_Rev1 already carries three sensors whose FHIR mapping is defined and ready. Five additional sensors are planned for future PCB revisions to expand clinical coverage — each with a specific recommended module for procurement.

Already on Band01_Rev1 PCB

MAX30102

Heart Rate & SpO2

Optical sensor using 660nm red and 940nm IR LEDs. Detects blood flow through the wrist to calculate HR (LOINC 8867-4) and blood oxygen saturation (LOINC 2708-6). Interface: I2C at address 0x57.



MPU-6050

Accelerometer & Gyroscope

6-axis IMU with $\pm 16g$ accelerometer and $\pm 2000^\circ/s$ gyroscope. Provides raw motion data for step counting (LOINC 55423-8), fall detection, and activity classification. Interface: I2C at 0x68.



MLX90632

Infrared Temperature

Non-contact IR thermometer measuring skin surface temperature to $\pm 0.2^\circ C$ accuracy. Calculates core body temperature via built-in DSP algorithm. FHIR: LOINC 8310-5. Interface: I2C at 0x3A.



Planned Future Hardware

AD8232

ECG

SparkFun DEV-12650. Single-lead ECG front-end. Detects arrhythmia and atrial fibrillation. LOINC 11524-6. Needs 2-3 skin electrodes.



Grove GSR

Stress / EDA

Seeed Studio SKU 101020052. Measures skin conductance for stress monitoring. LOINC 74767-6. Analog ADC interface.



ProtoCentral BP2

Blood Pressure

PPG-based wrist BP estimation. Systolic/Diastolic pressure. LOINC 55284-4. mm[Hg] units. High clinical demand.



TI AFE4490

Blood Glucose

Near-infrared optical front-end for non-invasive glucose monitoring. LOINC 15074-8. mmol/L. Emerging technology.



ICM-42688-P

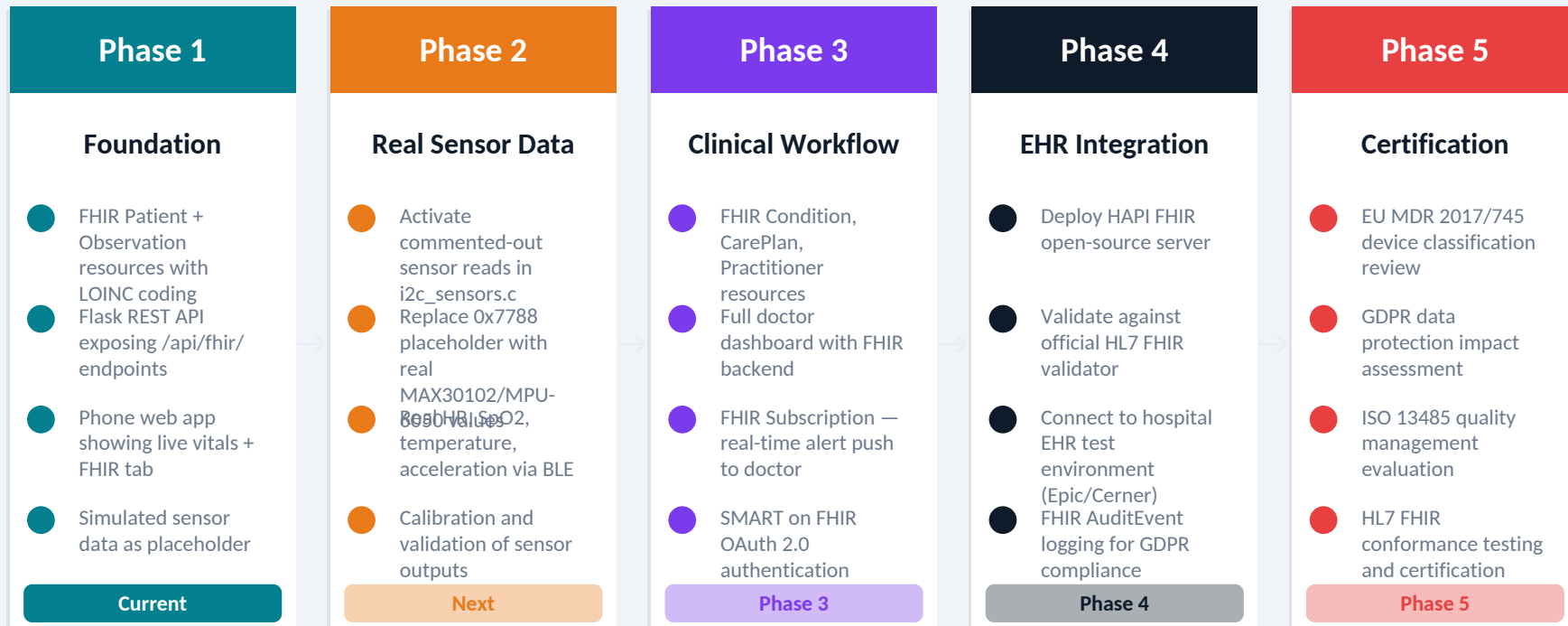
Respiratory Rate

TDK InvenSense high-precision IMU. Breathing rate derived from motion. LOINC 9279-1. /min. Can also be derived from MAX30102 PPG signal.



Five-Phase FHIR Implementation Roadmap

FHIR integration is built incrementally — each phase delivers real clinical value while setting up the next. Phase 1 is already underway. The roadmap is designed so the professor and project team can clearly see what comes next and why.



Data Privacy, GDPR Compliance & SMART on FHIR Authentication

Health data is the most sensitive category of personal data under GDPR. The Smartband V001 FHIR implementation must address privacy from the ground up — not as an afterthought. SMART on FHIR provides the industry-standard authentication layer.

GDPR & Regulatory Compliance

Encryption in transit

All FHIR API calls use HTTPS/TLS. Data in transit is never readable even if intercepted on the network. Mandatory under GDPR Article 32.

Encryption at rest

All patient data stored on the server is encrypted. The database itself is inaccessible without the decryption key. Required by GDPR Article 32.

Patient consent (Art. 7)

Every patient must give explicit consent before data collection begins. Consent is stored as a FHIR Consent resource and can be withdrawn at any time.

Right to erasure (Art. 17)

Patients can request deletion of all their health data. The FHIR DELETE operation removes all Observations, Conditions, and Patient resources permanently.

Audit logging

Every access and modification is logged as a FHIR AuditEvent resource. This creates a tamper-proof compliance trail required by ISO 27001 and HIPAA.

EU MDR 2017/745

The device must be classified under the EU Medical Device Regulation. FHIR compliance is increasingly considered during conformity assessment.

SMART on FHIR — OAuth 2.0 Auth Layer

SMART on FHIR is the authorization framework built on top of FHIR. It uses OAuth 2.0 tokens to control who can access which patient data. Without valid credentials, the FHIR API returns a 401 Unauthorized error — no data is exposed.



Doctor login

Receives a scoped token — can only see their assigned patients. Cannot access other doctors' patient lists.



Patient login

Receives a narrower token — read-only access to their own observations and care plan. Cannot modify records.



Research app

Requests anonymized, aggregated data only. Individual patient identifiers are never included in the response.



Token security

All endpoints require a valid OAuth 2.0 bearer token in the HTTP header. Tokens expire after a configurable period.

FHIR Server Options — Moving Beyond the Prototype

The current Flask implementation is a prototype. For production deployment — or even a serious academic demonstration — a dedicated FHIR server is required. The recommended option for this project is HAPI FHIR due to its open-source nature and full R4 compliance.

Why a dedicated FHIR server?

The Flask server is a custom implementation — it produces FHIR-shaped JSON but does not perform full FHIR validation, does not support FHIR search parameters, and does not implement the FHIR capability statement. A proper FHIR server handles all of this automatically and passes the official HL7 conformance tests.

HAPI FHIR

Open source Java · Free

The most widely used open-source FHIR server. Runs locally on a Pi or cloud VM. Full R4 compliance, built-in FHIR validator, and REST API. Recommended for this project.

★ Pick

Azure Health Data

Cloud FHIR service · Pay per use

Microsoft Azure's managed FHIR service. No server maintenance required. Integrates with Azure Logic Apps for automated workflows. Suitable for production cloud deployment.

AWS HealthLake

Cloud FHIR service · Pay per use

Amazon's FHIR-native health data lake. Includes built-in ML capabilities for predictive analytics. Best for large-scale research data aggregation across many devices.

Google Cloud FHIR

Cloud FHIR service · Pay per use

Google Cloud Healthcare API with FHIR R4 support. Integrates with BigQuery for advanced analytics. Best for research institutions already using Google Cloud infrastructure.

OpenMRS FHIR

Open source · Free

Open-source EHR with built-in FHIR module. Ideal for connecting Smartband data directly into a running EHR instance for academic demonstration purposes at HSHL.

Next Steps & Recommendations — Action Plan

1

Activate firmware sensor reads

Uncomment the sensor reading calls in `i2c_sensors.c` on the CC2640R2F firmware. Replace the `0x7788` placeholder value with real MAX30102, MPU-6050, and MLX90632 readings transmitted via BLE GATT.

2

Procure future sensors

Order: AD8232 ECG module (SparkFun DEV-12650), Grove GSR Sensor v1.2 (Seeed SKU 101020052), ProtoCentral BP2 blood pressure module. These three expand clinical coverage the most at lowest cost.

3

Expand the FHIR API

Add FHIR Condition resources for alert events, SpO2 Observation endpoint, and a `/api/fhir/observation/history` endpoint returning the last N readings as a FHIR Bundle with proper timestamps.

4

Deploy HAPI FHIR server

Install HAPI FHIR on the Raspberry Pi or a cloud VM. Migrate data storage from CSV to the HAPI FHIR database. This enables full FHIR search, validation, and capability statement support.

5

Implement SMART on FHIR authentication

Add OAuth 2.0 login to the web app and API. Use Keycloak or a similar identity provider. Role-based scopes — doctor vs patient — control what each user can see and modify.

6

FHIR Subscription for real-time alerts

Register FHIR Subscriptions so the doctor's phone receives a push notification the moment a patient's HR exceeds 100 BPM, temperature exceeds 38.5°C, or a fall is detected via acceleration.

7

FHIR conformance testing & EU MDR review

Run the official HL7 FHIR validator against all API endpoints. Commission an EU MDR 2017/745 device classification review to determine whether medical device certification applies.

The Bigger Picture — Why This Matters for Healthcare

FHIR is not just a technical standard — it is the infrastructure that makes personalized, data-driven medicine possible. The Smartband V001 project positions itself at the intersection of wearable IoT and clinical-grade health IT.

From Prototype to Clinical Tool

Today, the Smartband V001 is a student prototype — it collects simulated data and sends it to a phone app. With FHIR integration, the exact same hardware becomes a clinical-grade remote patient monitoring device whose data is readable by every hospital system in the world. The code and architecture described in this presentation are the bridge between those two realities.

Teaching Value of This Project

This project demonstrates in a single system: embedded firmware development (CC2640R2F), wireless communication (BLE), IoT gateway design (Raspberry Pi), RESTful API development (Flask), international health IT standards (FHIR R4, LOINC, UCUM), data privacy law (GDPR), and medical device regulation (EU MDR). It is a complete vertical slice of modern connected health.

168+

countries use LOINC codes

0 custom code

needed for EHR integration

FHIR R4

required by US ONC & EU MDR

5 sensors

already on Band01_Rev1 PCB

Epic • Cerner

both use FHIR R4 natively

18 references

in the FHIR prospects report

Thank You

Smartband V001 — FHIR Integration Prospects

- ◆ **HL7 FHIR R4** — The international standard — 168+ countries, required by Epic, Cerner, EU MDR, and US FDA
- ◆ **LOINC + UCUM** — Every sensor reading carries a universally understood code and unit — no ambiguity across systems
- ◆ **4-Layer Architecture** — Hardware → Firmware → FHIR Gateway → Clinical Consumers — each layer clearly separated
- ◆ **5-Phase Roadmap** — From current prototype to certified, EHR-integrated, GDPR-compliant clinical device
- ◆ **18 Academic References** — Grounded in peer-reviewed literature — Pinheiro et al. (2022) and HL7 FHIR specification